

Hazidien Ramadhan Utomo

PORTFOLIO



@Hazidien



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2025



ABOUT ME

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GIS SPECIALIST

HI I'M HAZIDIEN RAMADHAN UTOMO

I am a 23-year-old Urban and Regional Planning student at Institut Teknologi Sepuluh Nopember (ITS) with a passion for social interaction and exploring new opportunities. Known for my extroverted nature, strong critical thinking, collaborative spirit, and high self-esteem, I thrive in dynamic environments. I have strong skills in spatial analysis, remote sensing, and urban modeling, and I am proficient in tools such as ArcGIS, Google Earth Engine (GEE), and GeoSOS-FLUS. I use these tools to support land use modeling, urban planning strategies, and transportation analysis. My academic background is reinforced by hands-on experience in managing spatial data and conducting geospatial analysis to address urban challenges. I was also awarded 3rd place in a national-level GIS Analyst Infographic Competition organized by Universitas Pendidikan Indonesia, demonstrating my ability to merge technical expertise with effective visual communication.



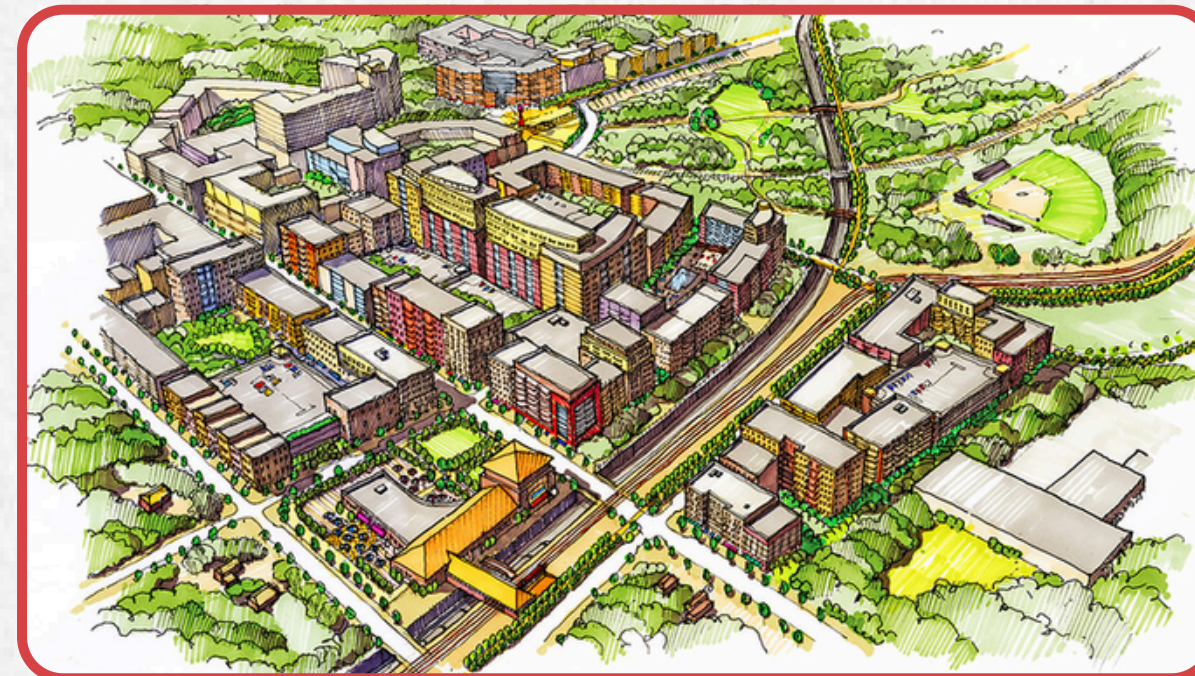


EDUCATION

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URBAN PLANNING

Urban and Regional Planning is a multidisciplinary field that focuses on designing and managing sustainable cities and regions. The program equips students with knowledge in spatial planning, land use, transportation, and urban design, supported by geospatial technology and data analysis.



- Urban Planning & Design Laboratory – Focuses on urban form, spatial planning, and urban design strategies.
- Regional Planning Laboratory – Explores regional development, policy, and spatial analysis at a wider territorial scale.
- Computation & Modeling Laboratory – Specializes in GIS, remote sensing, urban modeling, and transportation planning.





MY FOCUS

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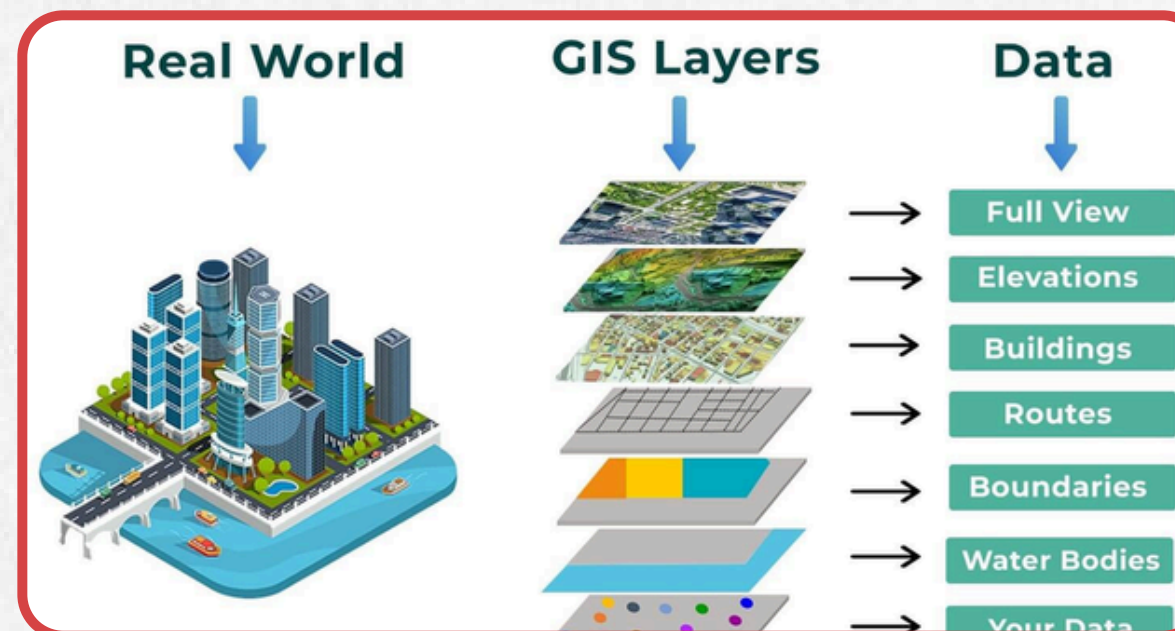
URBAN PLANNING STUDIO

Engaged in collaborative projects focusing on spatial planning, land use strategies, and urban design to create sustainable and livable cities.



GIS AND TRANSPORTATION ANALYST

Specialized in geospatial analysis, land use modeling, and transportation planning using tools such as ArcGIS, Google Earth Engine, and GeoSOS-FLUS.



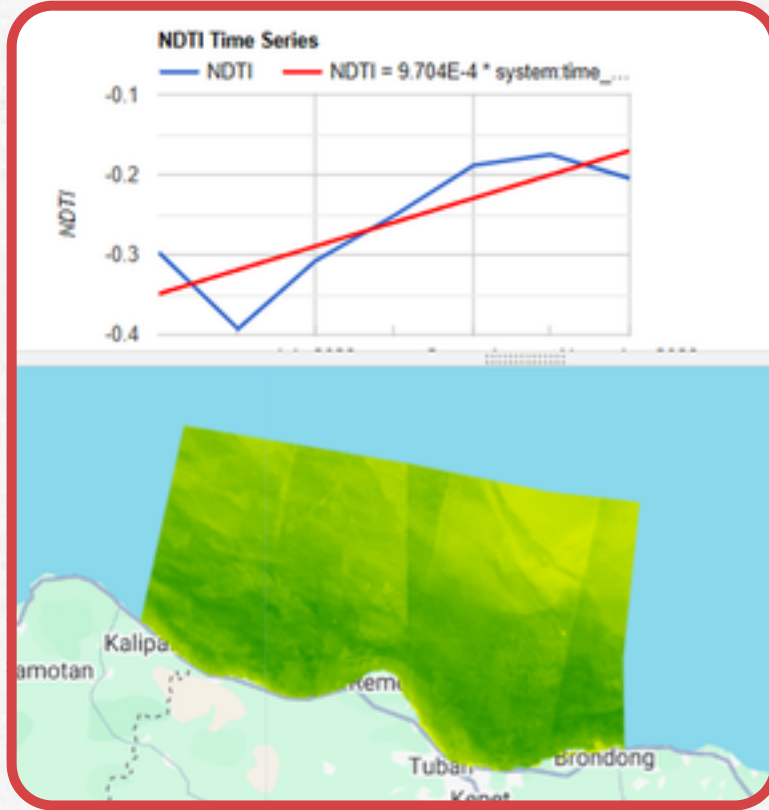


SKILL

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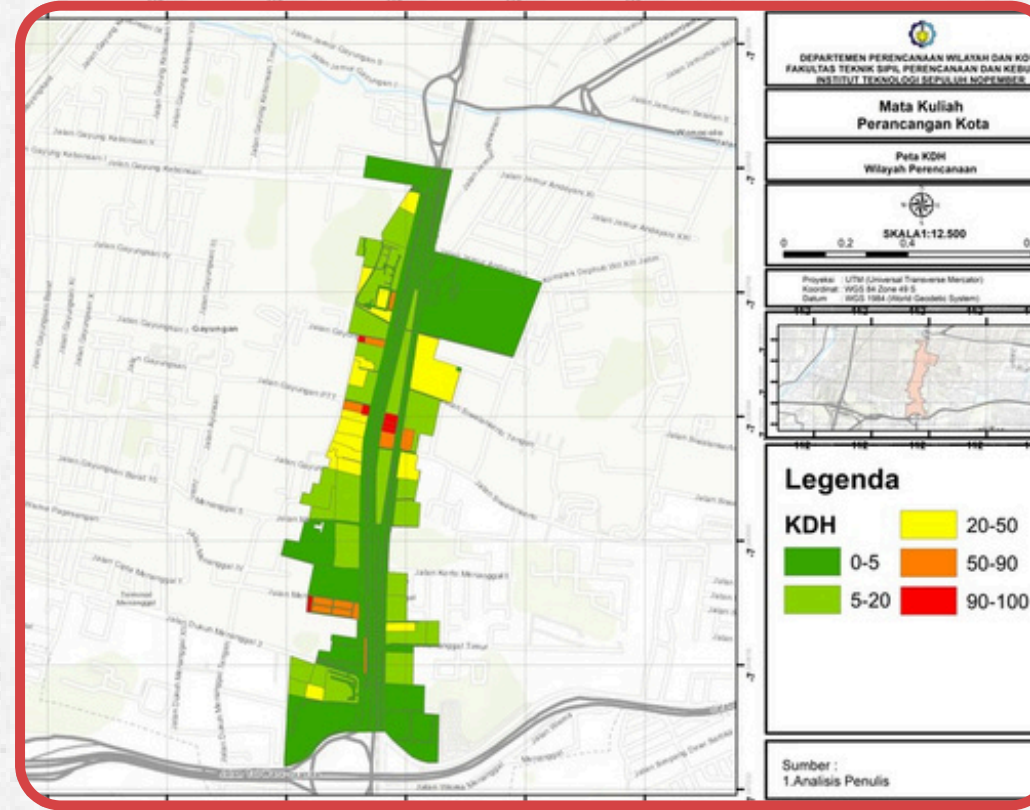
REMOTE SENSING

Conducting coastal and land cover analysis using Google Earth Engine (GEE) with indices such as NDTI, NDBI, NDWI, and others, as well as processing LiDAR data for spatial modeling and environmental monitoring.



GIS

Performing spatial data management and transportation analysis, including accessibility studies, land use modeling, and geospatial visualization using ArcGIS and other GIS tools.



URBAN DESIGN

Designing urban spaces and public areas with a focus on livability, walkability, and sustainability, integrating planning concepts into visual and practical design outputs.



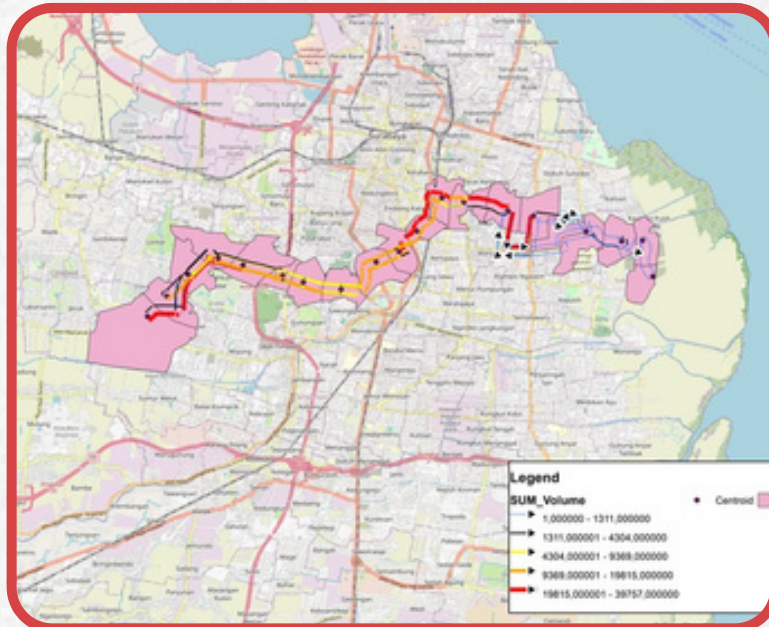


WORK EXPERIENCE

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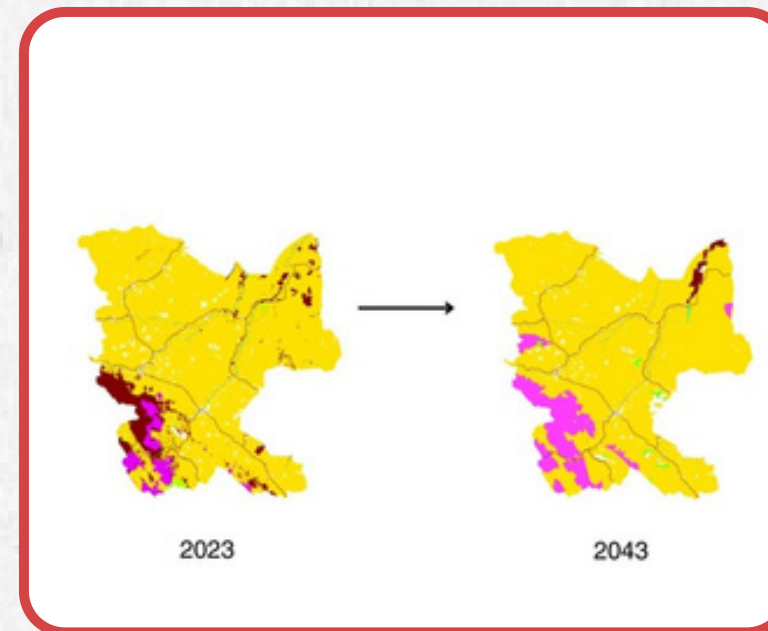
MRT SURVEYOR

Conducted household surveys and traffic counting to support urban mobility and transportation planning studies, including calculations using the Four Step Model (trip generation, trip distribution, mode choice, and route assignment).



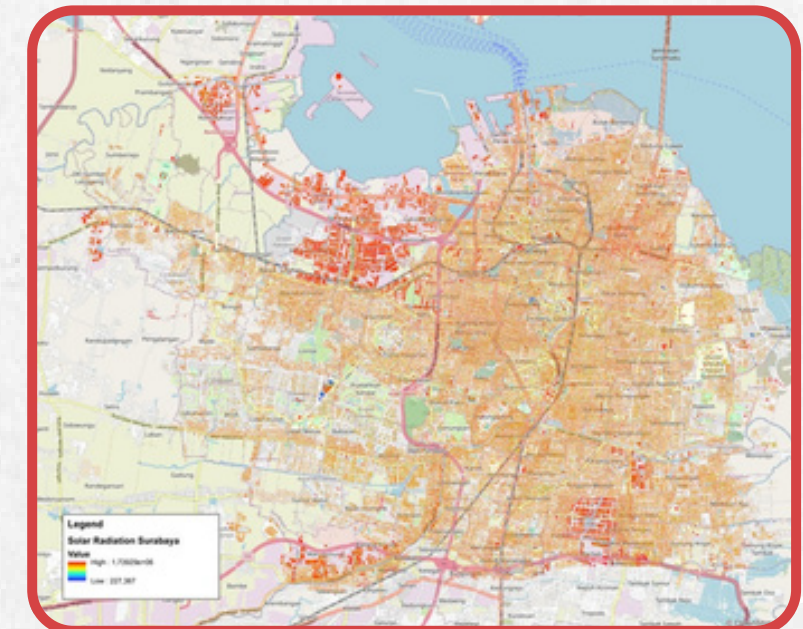
RESEARCH ANALYSIS

Performed land use change modeling using GeoSOS-FLUS in Rokan Hulu, focusing on spatial dynamics and future urban growth projection.



GREENOVA

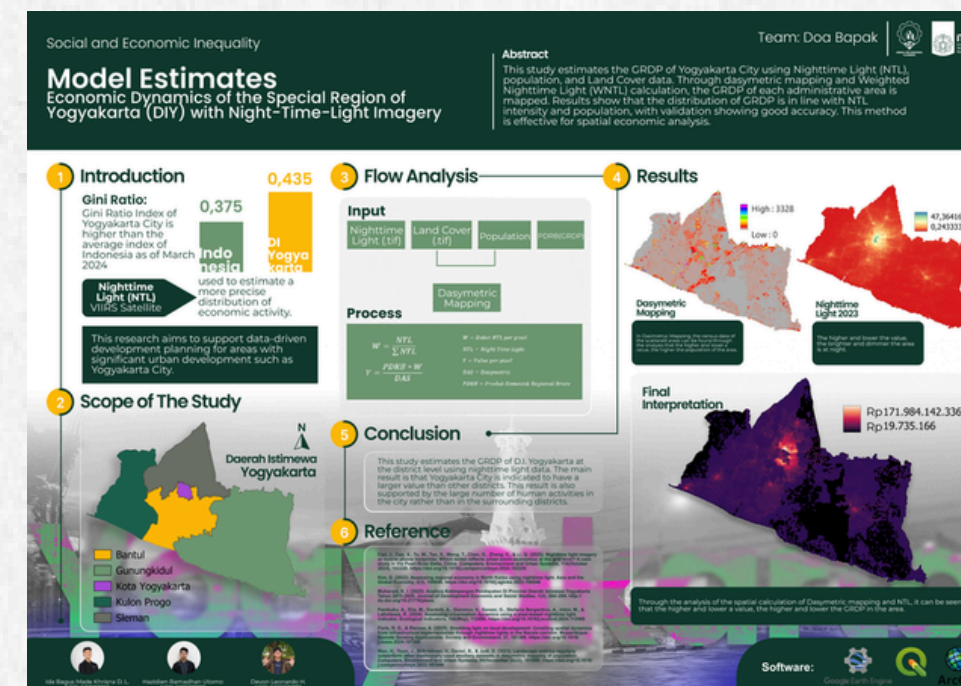
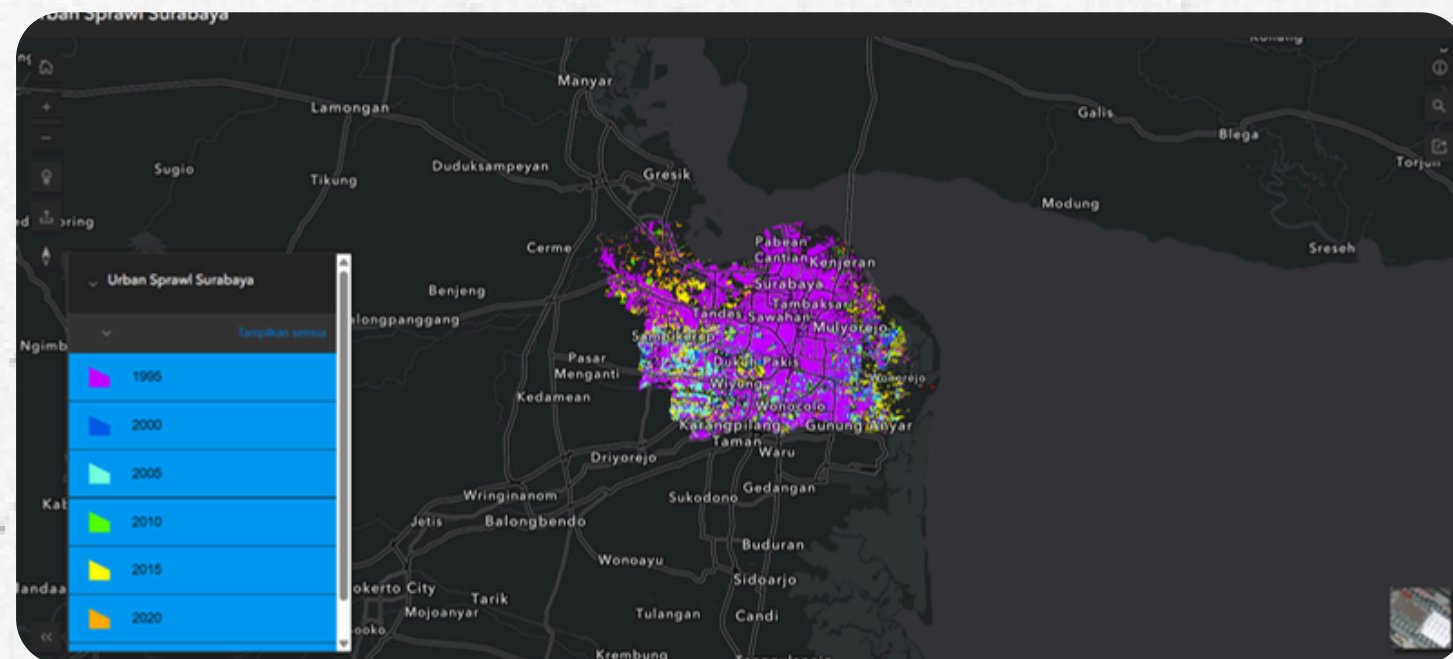
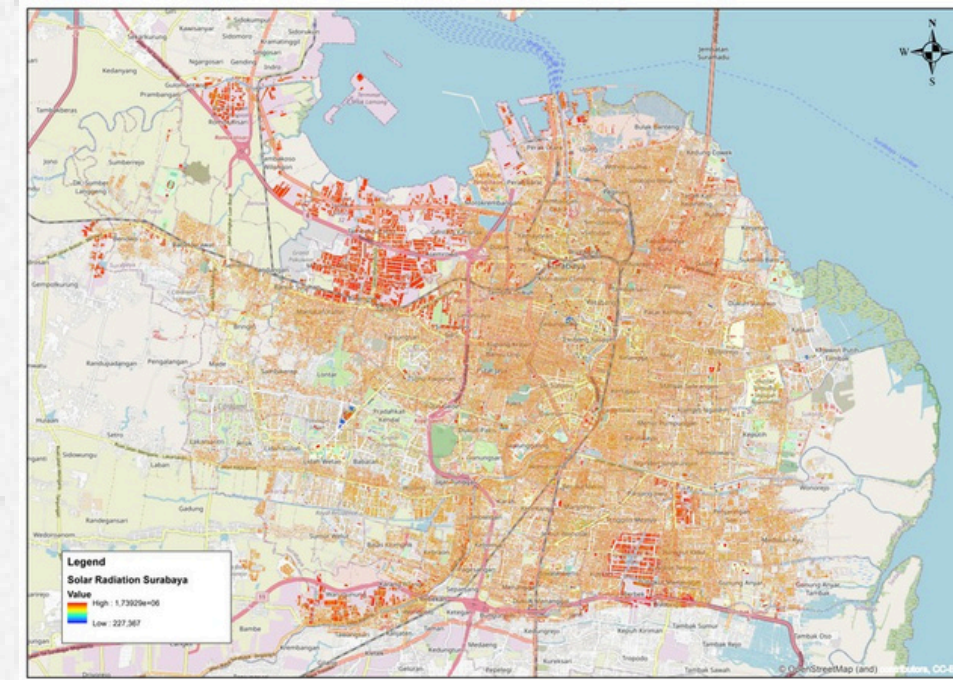
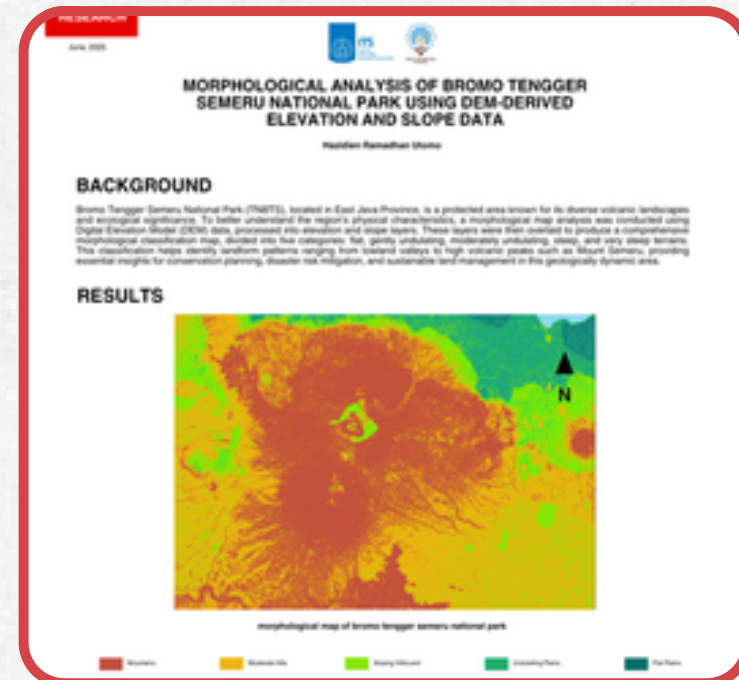
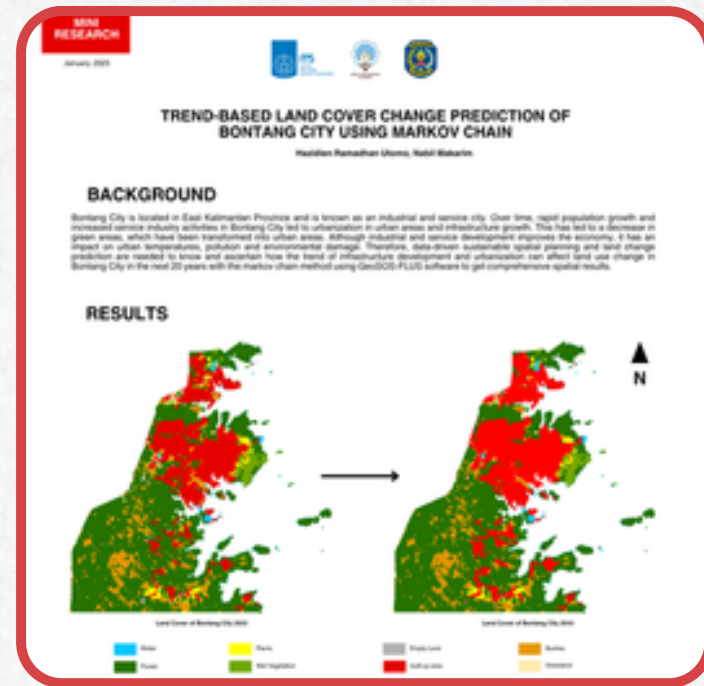
Assisted in spatial analysis for solar energy potential modeling in Surabaya by processing LiDAR data into nDSM to identify suitable areas for solar panel installation.





PROJECT PORTFOLIO

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PROJECT 1

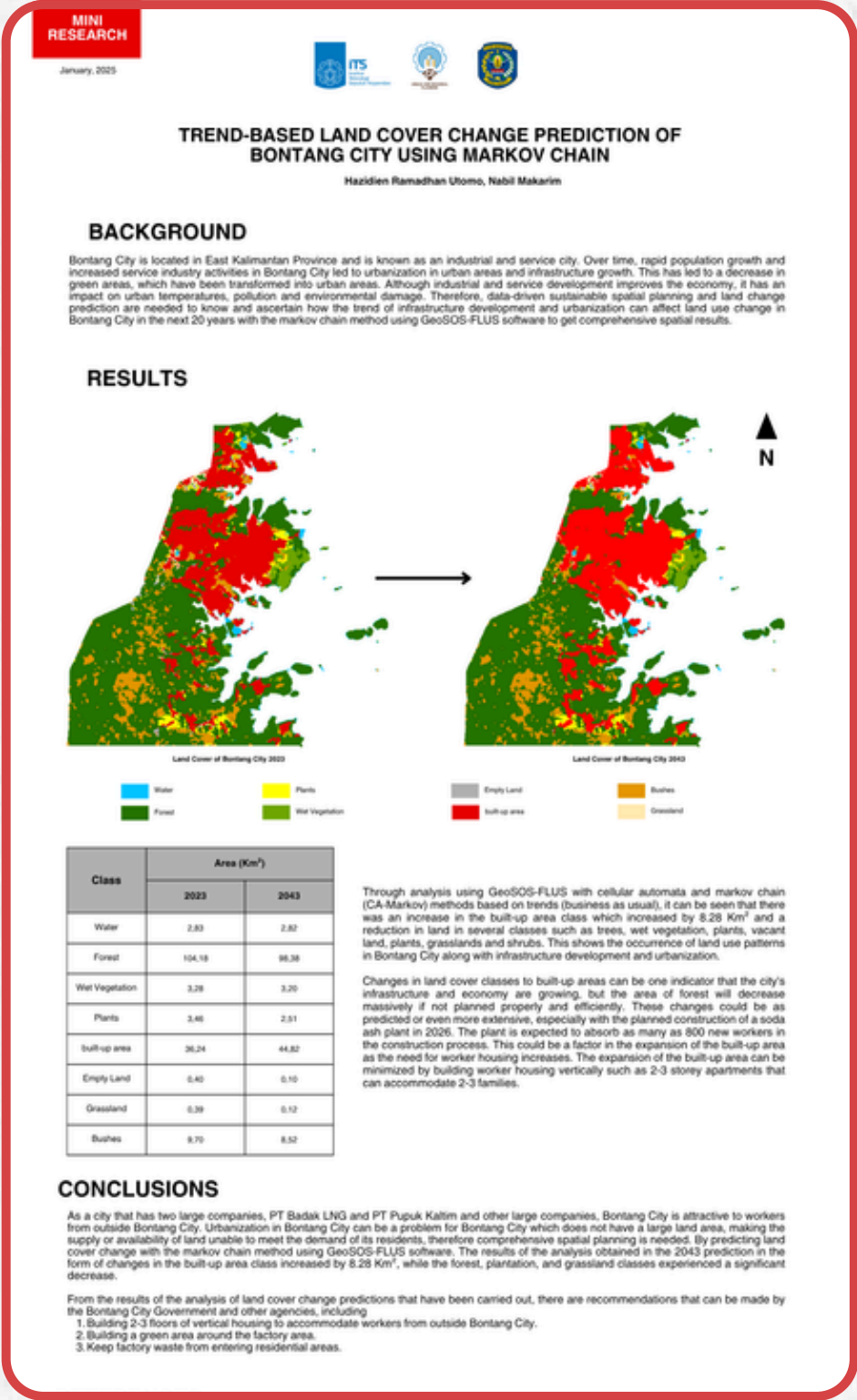
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LAND COVER CHANGES AT BONTANG CITY

This project predicts land cover change in Bontang City over the next 20 years using the Markov Chain method in GeoSOS-FLUS. Results show an 8.28 km² increase in built-up areas and significant loss of forests and vegetation due to urbanization and industrial growth. Recommendations include promoting vertical housing, preserving green areas, and improving waste management

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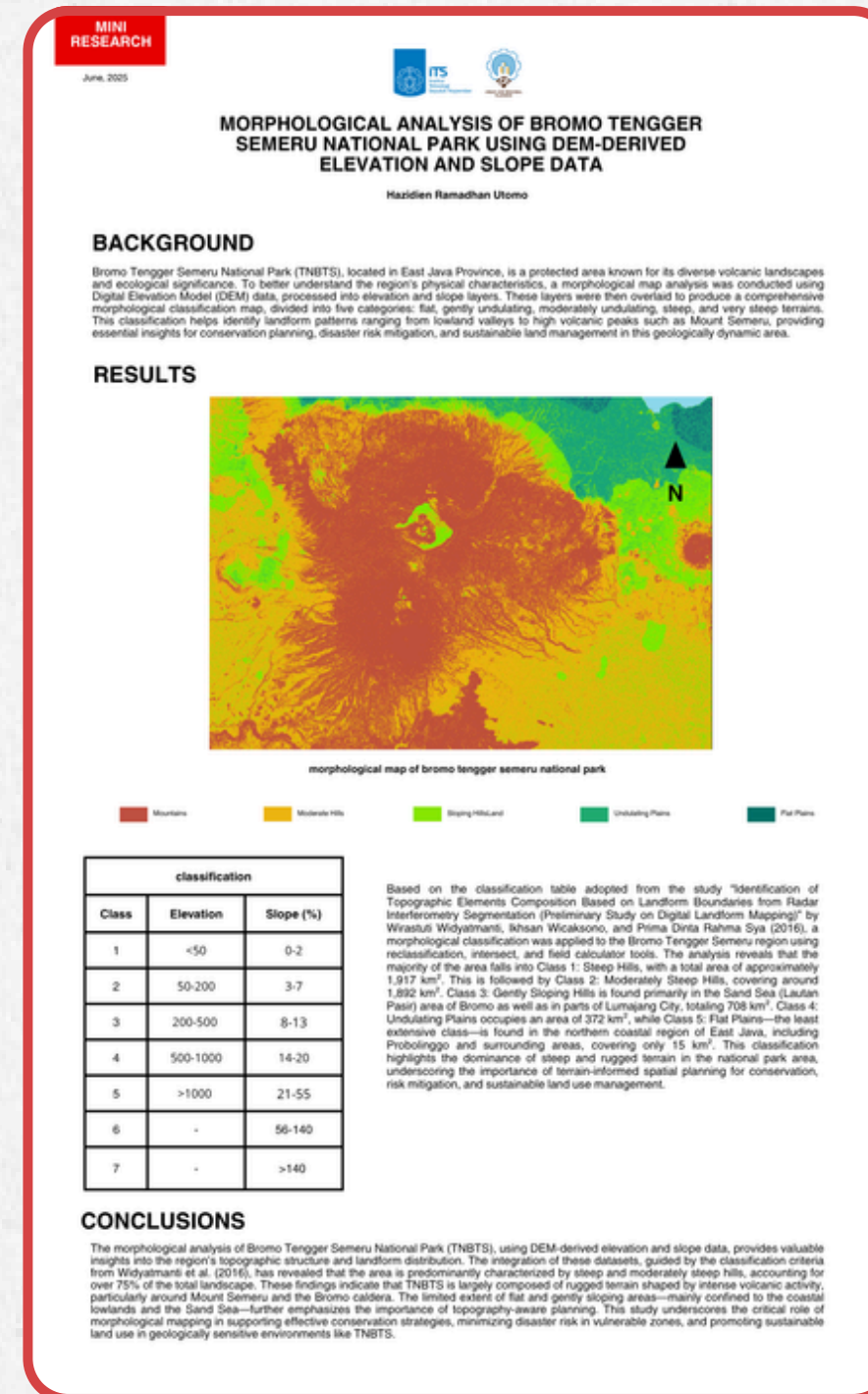
PROJECT 2

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MORPHOLOGICAL ANALYSIS OF BROMO TENGGER SEMERU NATIONAL PARK USING DEM-DERIVED ELEVATION AND SLOPE DATA

This study maps the morphology of Bromo Tengger Semeru National Park using DEM-derived elevation and slope data. Results show that over 75% of the area is dominated by steep and moderately steep terrain, shaped by volcanic activity around Mount Semeru and Bromo. These findings provide essential insights for conservation planning, disaster risk mitigation, and sustainable land management.

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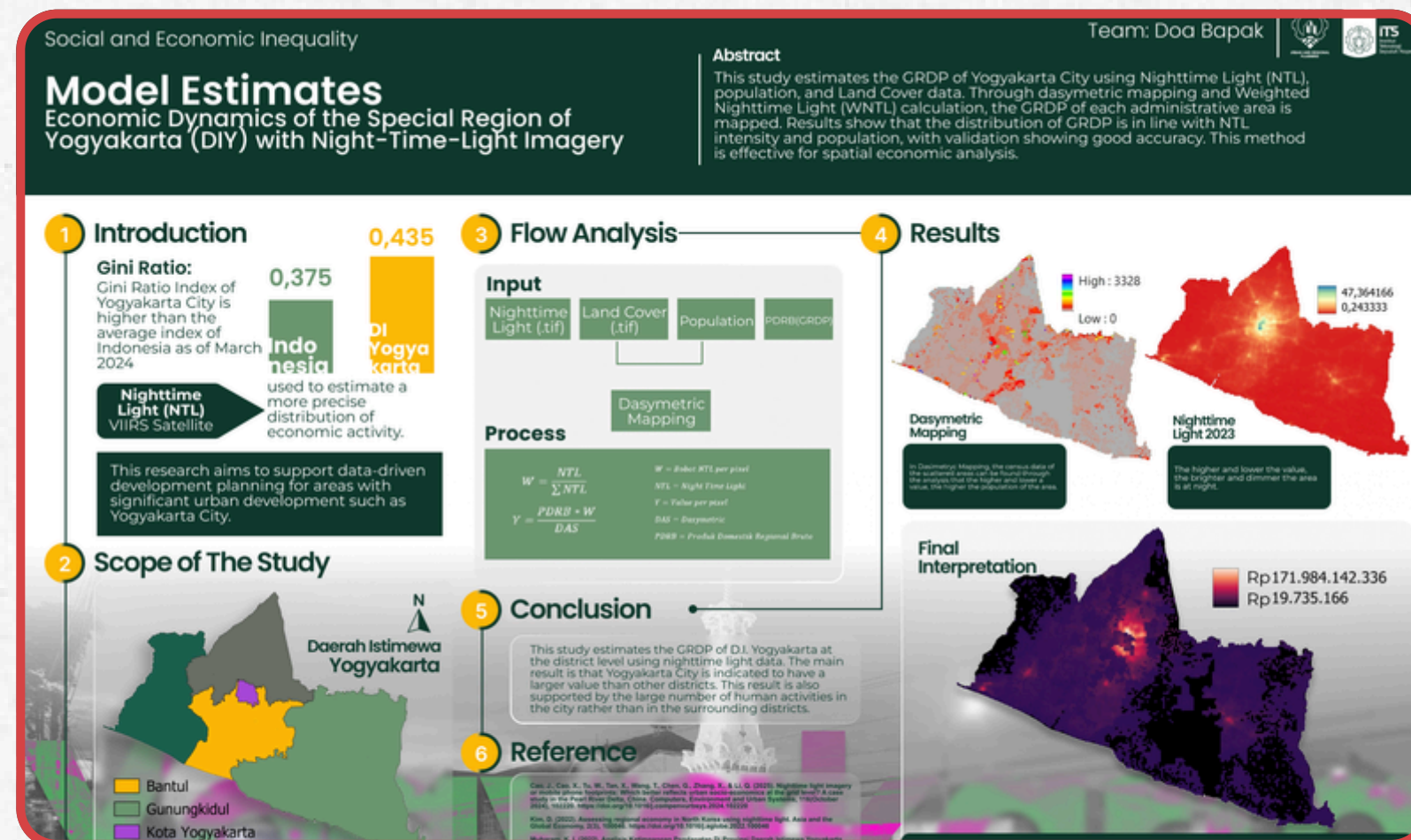


PROJECT 3

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MODEL ESTIMATES

ECONOMIC DYNAMICS OF THE SPECIAL REGION OF YOGYAKARTA (DIY) WITH NIGHT-TIME-LIGHT IMAGERY



This study estimates the GRDP of Yogyakarta Special Region using Night-Time Light (NTL) imagery, land cover, and population data with dasymetric mapping. Results show that Yogyakarta City has the highest economic activity compared to other districts, supported by its high urban intensity. The findings demonstrate that NTL is effective for spatial economic analysis and supports data-driven development planning.

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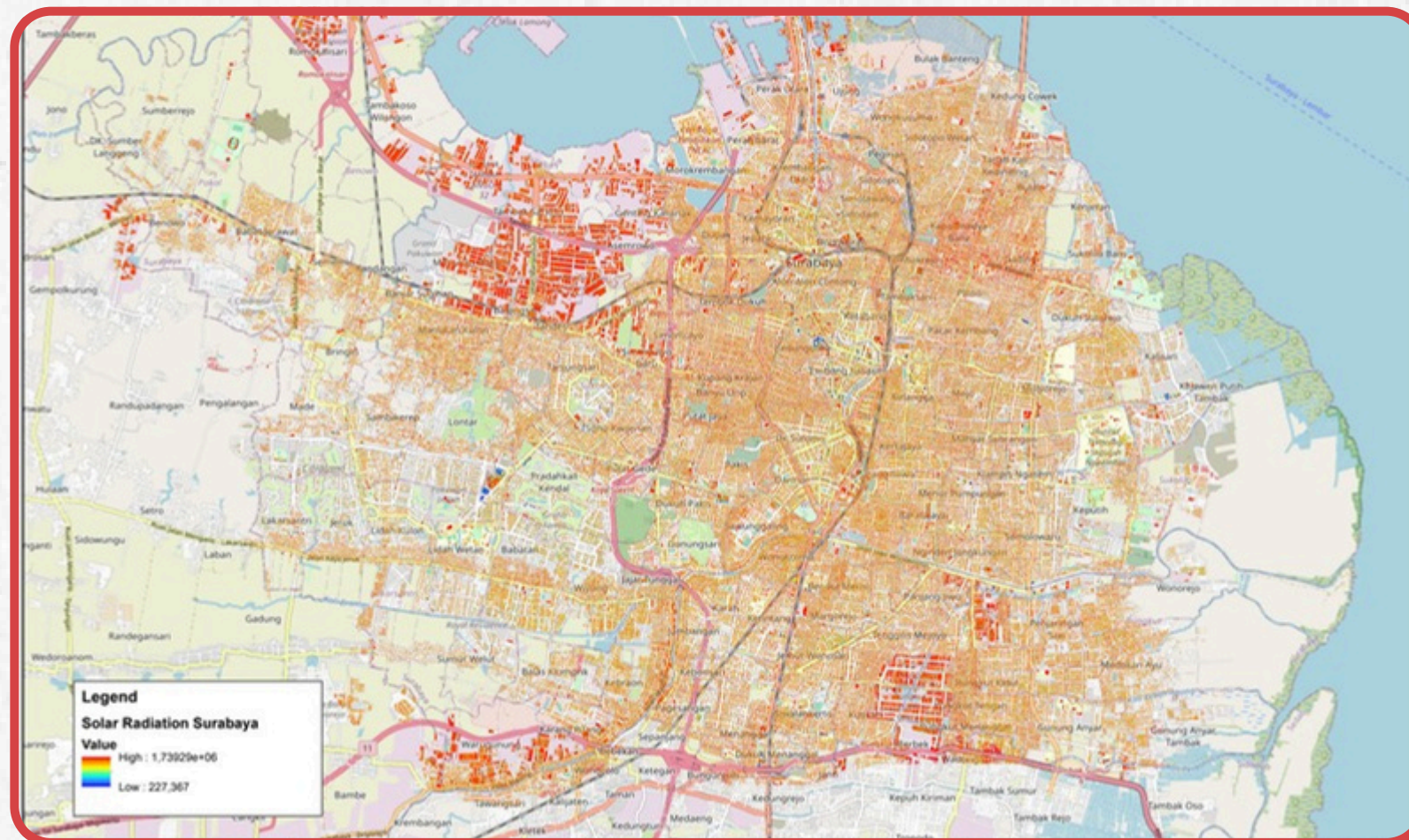




PROJECT 4

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SOLAR ENERGY POTENTIAL ANALYSIS IN SURABAYA USING LIDAR-DERIVED NDSM AND ARCGIS SOLAR RADIATION TOOL



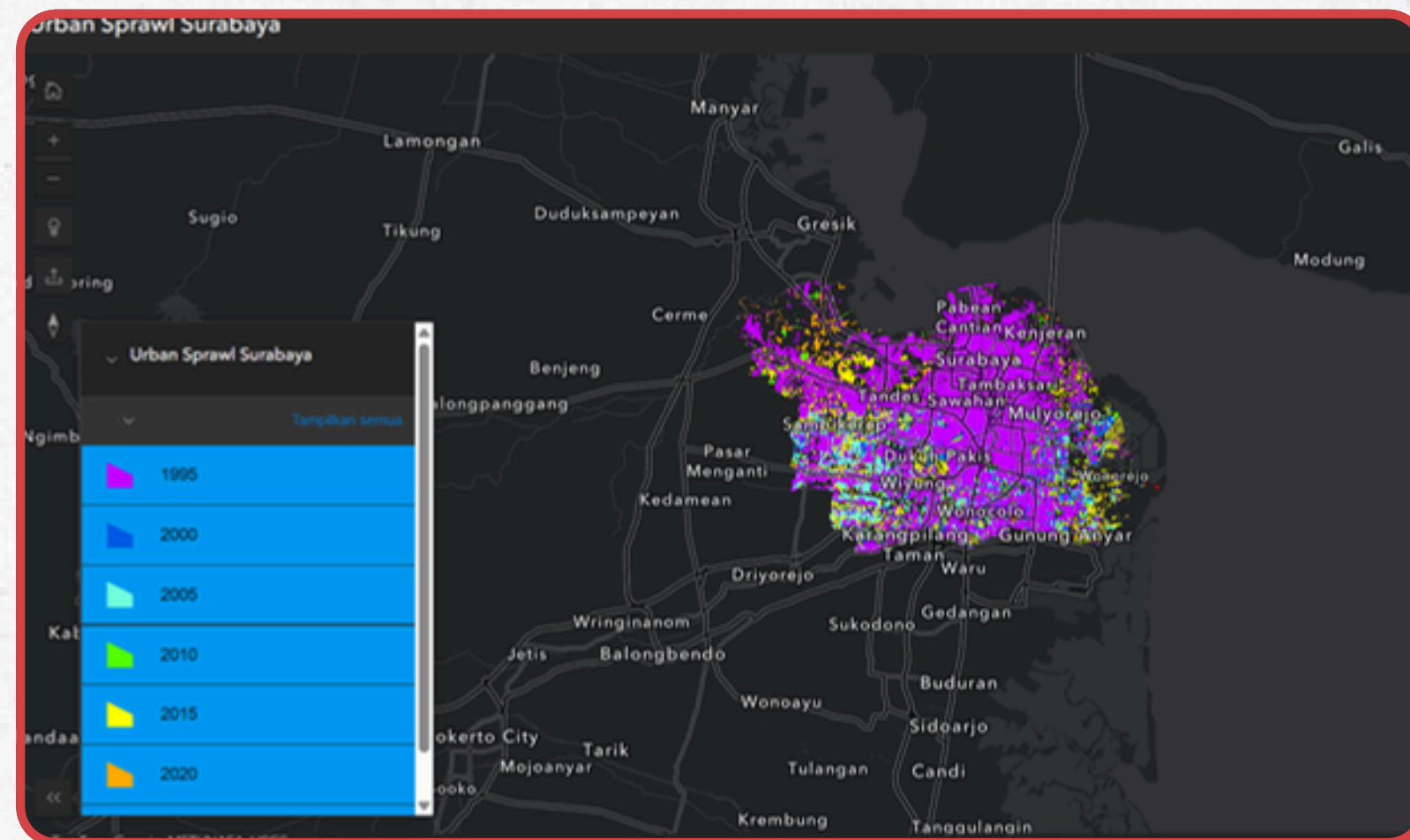
This project analyzes the solar energy potential of Surabaya using LiDAR-derived nDSM processed with the ArcGIS Area Solar Radiation tool, mapping annual rooftop solar potential and assessing spatial variability of solar irradiance ($\text{kWh/m}^2/\text{year}$) by considering building orientation, shading, land cover, and urban density, and further estimating rooftop photovoltaic electricity generation potential of up to approximately $234 \text{ kWh/m}^2/\text{year}$ per panel based on solar radiation data, module efficiency, and performance ratio assumptions.





PROJECT 5

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URBAN EXPANSION MAP AT SURABAYA CITY

This map shows the growth and distribution of population in Surabaya City from 1995 to 2025. The distribution of the population tends to point to the east and north. From this map, it is expected that stakeholders can make more appropriate policies in the future.

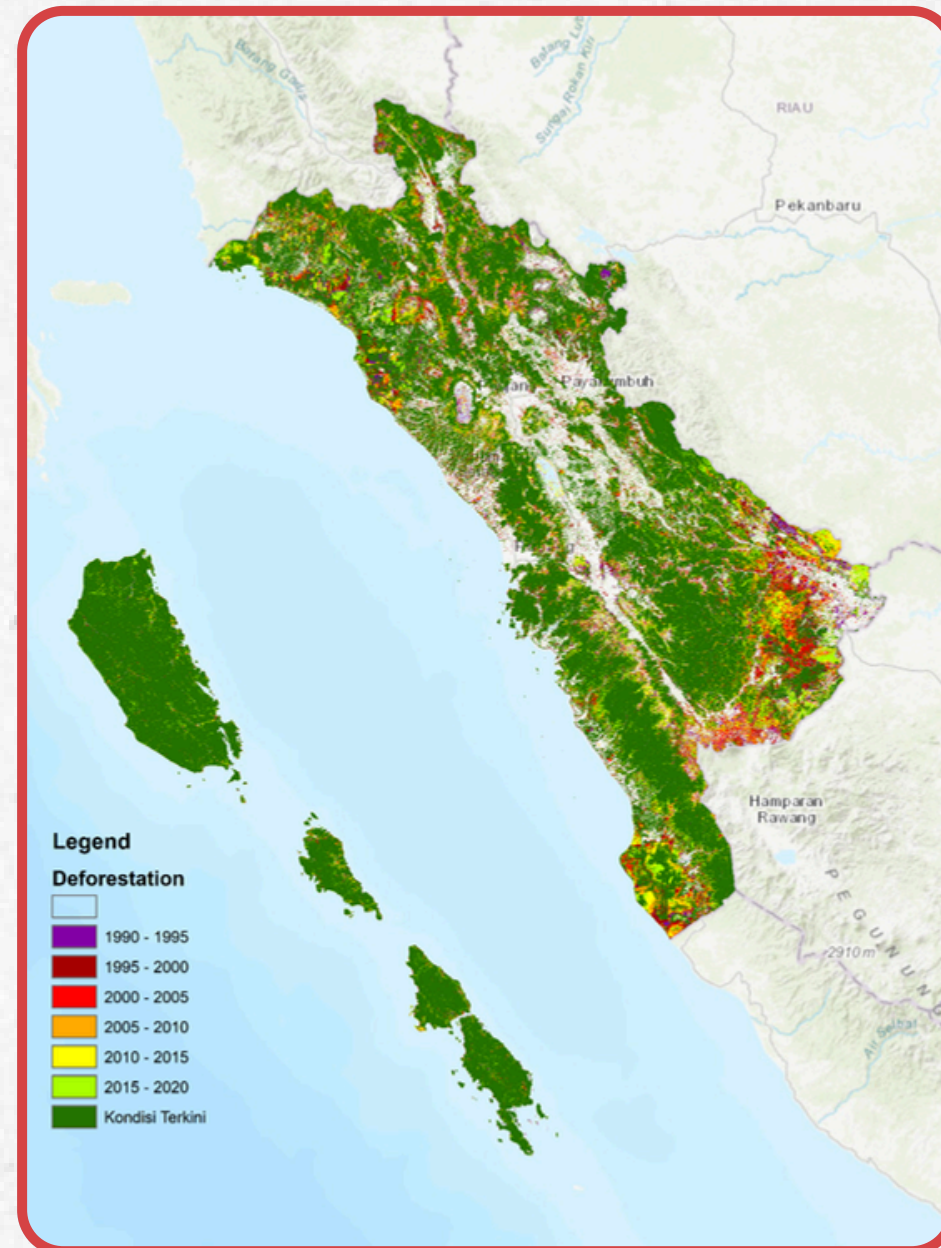
[CLICK HERE TO ACCESS INTERACTIVE MAP](#)





PROJECT 6

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SPATIO-TEMPORAL DEFORESTATION MODELING PROJECT – WEST SUMATRA

The Spatio-Temporal Deforestation Modeling Project in West Sumatra (Nov 2025–Dec 2025) involved analyzing multitemporal Landsat imagery from 2009 to 2024 using Google Earth Engine to identify patterns of forest loss, degradation, and regrowth, applying vegetation index-based thresholding and the Normalized Difference Fraction Index (NDFI) to detect deforestation dynamics, hotspots, and temporal change trajectories, and producing spatial outputs such as deforestation maps, hotspot analyses, and comparative method evaluations to support environmental monitoring and spatial planning decision-making.

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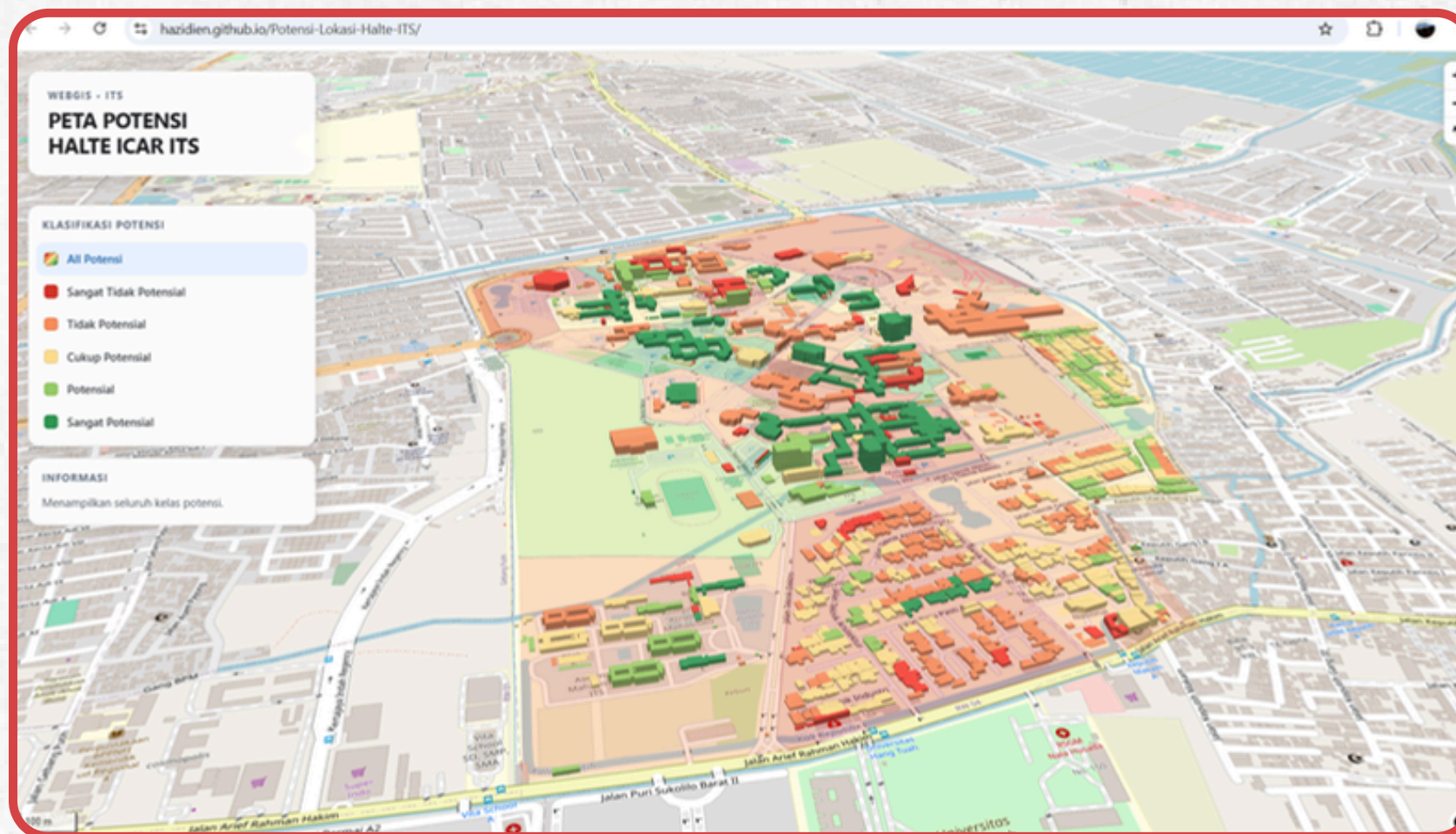




PROJECT 7

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SOLAR ENERGY POTENTIAL ANALYSIS IN SURABAYA USING LIDAR-DERIVED NDSM AND ARCGIS SOLAR RADIATION TOOL



Developed a 3D WebGIS using Urban Network Analysis (UNA) and Service Network Analysis (SNA) to identify buildings with high potential for campus transit stops across ITS, PENS, and PPNS. The analysis evaluates network accessibility and service coverage to support integrated campus transportation planning, with results visualized interactively through a 3D MapLibre WebGIS.

[click to acces](#)





PROJECT 8

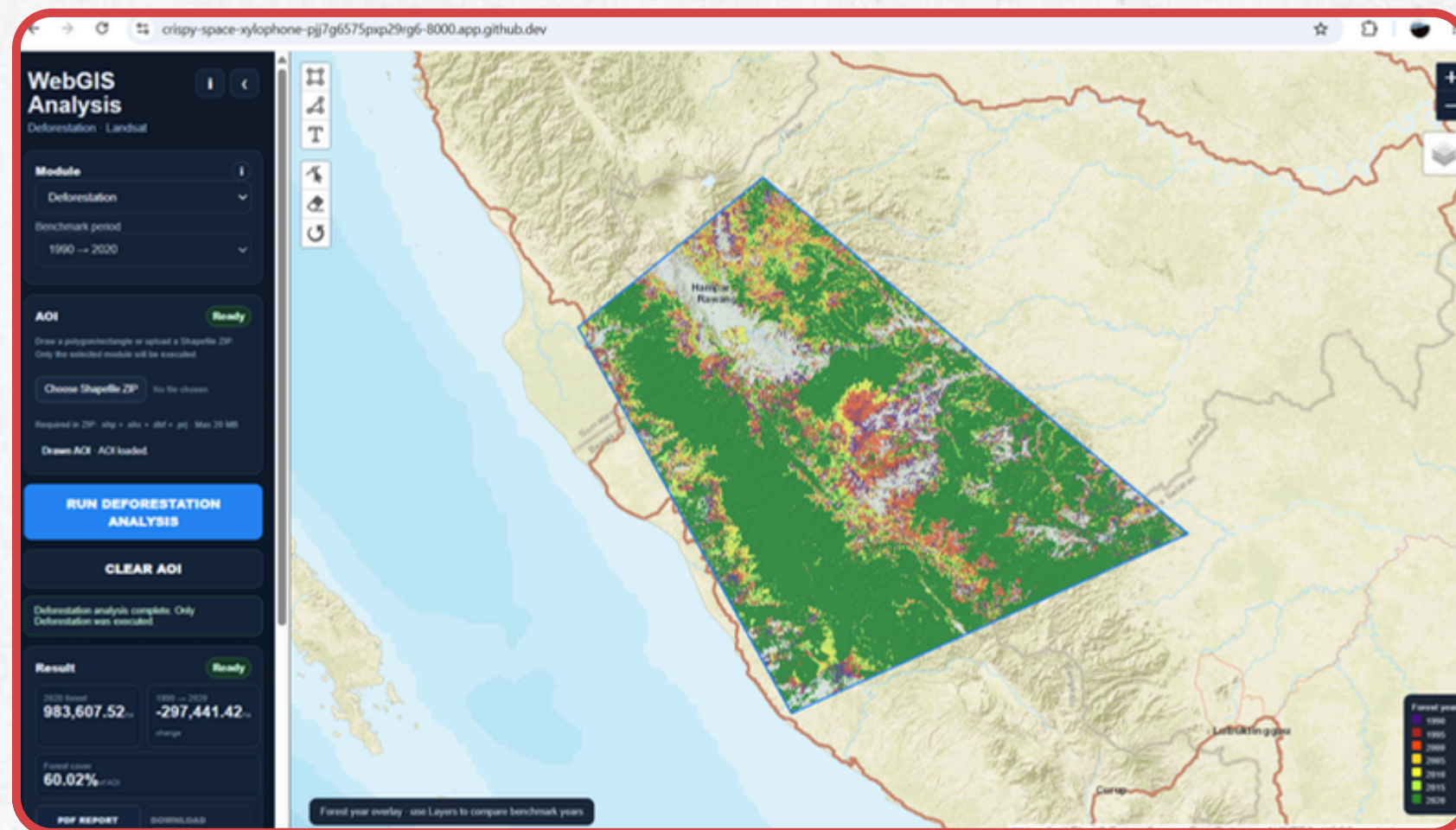
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DEFORESTATION MONITORING WEBGIS USING LANDSAT TIME-SERIES AND GOOGLE EARTH ENGINE



This project develops an interactive WebGIS for monitoring forest cover and deforestation using multi-temporal Landsat imagery and Google Earth Engine (GEE), featuring cloud masking, temporal compositing, forest classification, AOI-based analysis, time-series visualization, and exportable results from 1990–2020. The WebGIS was developed through prompt engineering and AI-assisted coding using GPT-based AI, with the workflow, requirements, and system design directed and refined by the developer.

Demo: Currently available as a demonstration through GitHub Codespaces and requires the backend service to be active. The application is not continuously hosted. Please contact me if you would like to test the demo. **[Click to Open WebGIS Demo](#)**

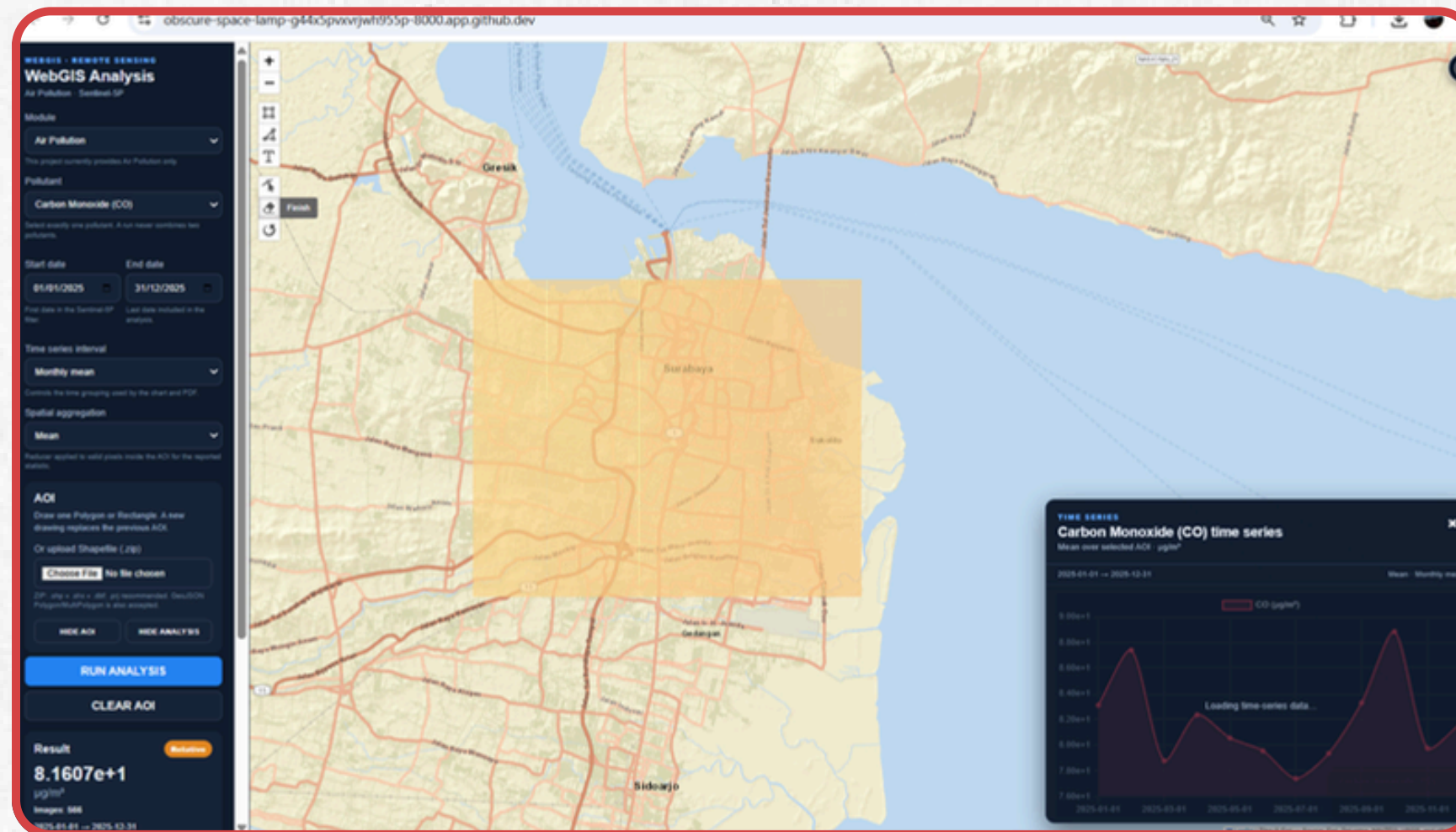




PROJECT 9

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INTERACTIVE AIR POLLUTION WEBGIS USING SENTINEL-5P AND GOOGLE EARTH ENGINE



WebGIS for analyzing atmospheric pollution using Sentinel-5P and Google Earth Engine. Users can select pollutants, define dates and AOI, generate spatial concentration maps, statistics, and time-series visualizations. The WebGIS was developed through prompt engineering and AI-assisted coding using GPT-based AI, with the workflow, requirements, and system design directed and refined by the developer.

Demo: Currently available as a demonstration through GitHub Codespaces and requires an active backend service. The application is not continuously hosted. Please contact me if you would like to test the WebGIS.

→ **[Click to Open WebGIS Demo](#)**



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LET'S WORK
TOGETHER!



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